STEM Sentence Writing Puzzles

Course: Technical & Scientific Writing in English | Target: University Science & Engineering

Student Name: _____ **Student ID:** _____ **Date:** _____

Puzzle 1: Subject & Predicate Breakdown (Biotech & AI)

SYNTAX DIVISION

Task: (1) Draw a vertical line (|) separating the **Complete Subject** from the **Complete Predicate**. (2) Underline the **Simple Subject** core. (3) Circle the **Main Verb / Auxiliary**.

A. The synthesis of high-purity graphene oxide nanosheets for battery electrodes requires a controlled oxidation process.

B. Training deep neural networks on uncurated medical imaging datasets increases the risk of algorithmic bias.

C. Among the various renewable energy storage solutions, green hydrogen fuel cells remain the most promising option.

Puzzle 2: Relative Clauses in Technical Specifications

DEFINING VS. NON-DEFINING

Task: Combine each pair of technical observations into one precise sentence using relative pronouns (*which / that / who*) and proper comma punctuation.

1. Defining (Essential Restriction):
Statement A: The semiconductor chips were rejected during quality assurance testing.
Statement B: These chips contained microscopic structural micro-cracks.

2. Non-Defining (Supplemental Context):
Statement A: The James Webb Space Telescope operates at Lagrange Point 2.
Statement B: The James Webb Space Telescope detects faint infrared radiation from early galaxies.

Puzzle 3: Scientific Register & Sentence Function

SCIENTIFIC FUNCTIONS

Base Technical Finding: *"The lithium-ion battery overheated during rapid charging cycles."* Rewrite into three distinct formal technical registers.

A. Formal Interrogative (Hypothesis / Research Question):

B. Technical Imperative (Experimental Procedure Directive):

C. Formal Academic Exclamatory (Emphatic Scientific Claim - No informal words):

Puzzle 4: Technical Sentence Architecture

STRUCTURAL SYNTHESIS

Given Scientific Blocks:
[1] *Although quantum computers process complex algorithms exponentially faster...*
[2] *Qubits remain extremely susceptible to environmental noise...*
[3] *Engineers must develop robust error-correction protocols...*

A. Complex Sentence (CX) using Block 1 & Block 2:

B. Compound-Complex Sentence (CCX) combining Blocks 1, 2, and 3:

Instructor Answer Key & Technical Notes

Course: Technical & Scientific Writing in English | For Evaluation and Grading Guidance

Markup Key for Puzzle 1: Underlined = Simple Subject Core | **Boxed** = Main Verb/Auxiliary | | = Subject/Predicate Division Boundary.

Puzzle 1 Key: Subject & Predicate Breakdown

ANSWER KEY

A. The synthesis of high-purity graphene oxide nanosheets for battery electrodes | **requires** a controlled oxidation process.

B. Training deep neural networks on uncurated medical imaging datasets | **increases** the risk of algorithmic bias.
Note: "Training..." is a gerund phrase acting as a singular noun subject.

C. [Among the various renewable energy storage solutions,] green hydrogen fuel cells | **remain** the most promising option.
Note: Prepositional phrase introduces the statement; complete subject begins with "green hydrogen...".

Puzzle 2 Key: Technical Relative Clauses

ANSWER KEY

1. Defining Clause (No commas; restrictive scope):

"The semiconductor chips **that contained microscopic structural micro-cracks** were rejected during quality assurance testing."
(Technical Note: Crucial distinction—only the damaged chips were rejected, not all chips.)

2. Non-Defining Clause (Commas required; supplemental facts):

"The James Webb Space Telescope**, which detects faint infrared radiation from early galaxies,** operates at Lagrange Point 2."
(Technical Note: Use "which" preceded by a comma because the proper noun is already unique.)

Puzzle 3 Key: Scientific Functions & Registers

ANSWER KEY

A. Formal Interrogative (Research Question):

"Under what thermal conditions does rapid charging induce thermal runaway in lithium-ion batteries?"

B. Technical Imperative (Procedure Directive):

"Monitor core cell temperature continuously during rapid charging cycles to prevent thermal runaway."

C. Formal Academic Exclamatory (Emphatic Scientific Claim):

"How unexpectedly severe the thermal degradation observed in the lithium-ion cell during rapid charging was!"
OR: "Rarely has a battery cell exhibited such rapid thermal instability under standard charging protocols!"

Puzzle 4 Key: Technical Sentence Architecture

ANSWER KEY

A. Complex Sentence (CX = 1 Dependent + 1 Independent):

"Although quantum computers process complex algorithms exponentially faster, qubits remain extremely susceptible to environmental noise."

B. Compound-Complex Sentence (CCX = 1 Dependent + 2 Independent):

"Although quantum computers process complex algorithms exponentially faster, qubits remain extremely susceptible to environmental noise, so engineers must develop robust error-correction protocols."